

Technical Datasheet — VBF Power-Tool Cell (V4 Margin-fix)

Case: t3_r1_flash | Generation date: 2026-08-25 | Doc: VBF-T3R1FLASH-DSH-01

Field	Value	Source
Rated capacity	3.44 Ah (nominal 3.438 Ah simulated at 1C, DFN)	r2_v4_1c_dfn.json:capacity_ah
Nominal voltage	midpoint 3.88 V	r2_v4_energy.json:midpoint_voltage_v
Voltage window	2.5 – 4.2 V	parameter set
Rated energy	12.42 Wh	r2_v4_energy.json:energy_wh ($\int V \cdot I \, dt$)
Gravimetric energy density	410 Wh/kg	r2_v4_energy.json (contract caliber; electrolyte/casing excluded)
Volumetric energy density	864 Wh/L	r2_v4_energy.json
Power density (theoretical peak, $P=V^2/4R$)	111,807 W/kg	r2_v4_energy.json:power_density_w_kg
DC resistance (1C, first 10% time)	1.26 mΩ	r2_v4_energy.json:dcr_ohm
Max continuous discharge rate	5C with 98.7% capacity retention	r2_v4_5c_dfn.json + derived
Fast-charge capability (4C @45 °C)	CC-phase no lithium plating (anode min +22.6 mV); T_max 48.8 °C	r2_v4_4c_dfn.json
4C CC-phase charge acceptance	0.617 Ah before 4.2 V cutoff (CV would complete; honest note)	r2_v4_4c_dfn.json:capacity_ah
Operating temperature range	Per simulation: charge tested at 45 °C ambient; discharge at 25 °C	protocol conditions (honest: full range not characterized)
Cycle life	Not simulated (no aging criterion in task; requires aging model run)	honest annotation
Safety determination	Plating: NO (anode potential > 0 V throughout 4C charge); T_max 48.8 °C < 60 °C limit	r2_v4_4c_dfn.json
Dimensions	65 mm × 1580 mm strip, stack 0.14 mm (shell not modeled)	parameter set + calc-energy
Mass	30.30 g (contract layer mass; +5.5 g electrolyte estimate = 35.8 g)	r2_v4_energy.json:mass_kg
Chemistry	NMC-type cathode / graphite anode; high-transport LiPF6-class electrolyte (σ 2.4 S/m, t_{eff} 0.5)	params_r2_v4.json